

Lecture 10: Testing Single Equations

Last lecture, we studied how to test hypotheses of the form $\beta = \beta_0$, where β is a (single) coefficient in our regression. We showed that, under our standard three assumptions about OLS,¹ the t-statistic, $\frac{\hat{\beta} - \beta_0}{\widehat{SE}(\hat{\beta})}$, is asymptotically standard normal, meaning that it is distributed $N(0, 1)$ in large samples. This allowed us to do hypothesis tests and construct confidence intervals: we just needed to compare the t-statistic (or rather the absolute value of the t-statistic) under the null to the critical value coming from the standard normal.

Today, we will extend this theory to tests of single (linear) equations. These are equations that can be expressed as $\theta = \theta_0$, where θ is a linear combination of regression coefficients. We will show how our earlier results extend to the single equation case, and how we can obtain the necessary estimates of $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$.

Grants to Microenterprises: de Mel, McKenzie, and Woodruff (2008)

For today's lecture, we will use de Mel, McKenzie, and Woodruff's 2008 study of grants to microenterprises as a running example. An important quantity in the study of firms is the return to capital, which is especially important in developing countries, where it is believed that returns to capital are high, particularly for microenterprises. However, studying the returns to capital is challenging because capital tends to be highly non-random. In fact, theory tells us that in a well-functioning market, the firms with the most capital will also be the firms with the highest productivity, which creates a natural omitted variable bias problem.

To solve this problem, de Mel et al. ran a randomized controlled trial among microenterprises in Sri Lanka. After a baseline survey, they randomly assigned some microenterprises to receive grants and others to be in the control group. Within the treated group, the microenterprises were randomly assigned to one of four treatments: they could receive a small grant (10,000 Sri Lankan Rupees) or a large grant (20,000 Sri Lankan Rupees), and the grant could be either in cash or in-kind (for the in-kind grant, the research team asked the microenterprise what sort of capital goods they wanted for their business, and gave them the grant in the form of those goods). They ran their experiment for nine waves, collecting data on profits both before and after the grants were rolled out. By randomizing, de Mel et al. were able to get around the omitted variable bias problem, since randomization ensures that the treatment is uncorrelated with the error term (in large samples).

To estimate the effect of each treatment, they run the following regression:²

$$\text{Profit}_{it} = \alpha + \beta_{LC} \text{Large Cash Grant}_{it} + \beta_{SC} \text{Small Cash Grant}_{it} \\ + \beta_{LIK} \text{Large In-Kind Grant}_{it} + \beta_{SIK} \text{Small In-Kind Grant}_{it} + \gamma_t + \varepsilon_{it}$$

where Profit_{it} is firm i 's profits in month t , $\text{Large Cash Grant}_{it}$ is an indicator that is equal to one if the firm is assigned to receive a large cash grant, and zero otherwise, and the other grant indicators are defined equivalently.³ Finally, note that γ_t denotes a set of dummy variables for each month: for now we will not pay any attention to these, but we will discuss them more in future lectures on panel data.

There are many questions we might be interested in asking in this setting, which we can formulate as hypotheses to test. Before estimating the return to capital, de Mel et al. are interested in testing whether cash grants have the same effects as in-kind grants, and also in testing whether small grants are half as effective as large grants. In general, these sorts of hypotheses involve comparing multiple coefficients (e.g. we might want to test if $\beta_{SC} = \frac{1}{2}\beta_{LC}$), and so we will need to develop our theory a bit more to test these hypotheses.

¹These assumptions are no perfect collinearity, IID sampling, and the orthogonality condition.

²De Mel et al. actually also include firm fixed effects (dummy variables for each firm) in their main specification, although it is not necessary in their setting and has little effect on the results.

³In the first month, the indicator is zero for all firms, then it becomes one for the treated firms starting in the month they receive the grant.

Linear Combinations of Coefficients

To help us test these sorts of hypotheses, we will define a new term, θ , which is a linear combination of the OLS coefficients. It turns out that for any linear equation that we want to test, we can define our combined parameter, θ , and then we can write the null hypothesis as $\theta = \theta_0$. For example, suppose that we wish to test whether cash grants and in-kind grants have the same effect on profits. Imagine that we run a simplified version of the earlier regression (pooling the small and large grants of each type):

$$\text{Profit}_{it} = \alpha + \beta_C \text{Cash Grant}_{it} + \beta_{IK} \text{In-Kind Grant}_{it} + \gamma_t + \varepsilon_{it}$$

Here, our null hypothesis is $\beta_C = \beta_{IK}$. We can rewrite this by defining $\theta = \beta_C - \beta_{IK}$, and saying that our null hypothesis is $\theta = 0$.

We can also use this technique to test whether small grants (which are half the size of large grants) have half the effect on profits. This time, imagine we run the following simplified regression:

$$\text{Profit}_{it} = \alpha + \beta_L \text{Large Grant}_{it} + \beta_S \text{Small Grant}_{it} + \gamma_t + \varepsilon_{it}$$

where now we have pooled cash and in-kind grants, and instead focus on the grant size. Here, our null hypothesis is $\beta_S = \frac{1}{2}\beta_L$, which we can also write as $\theta = 0$, where $\theta = \beta_S - \frac{1}{2}\beta_L$. We can also use this technique to test other (linear) hypotheses, such as the *average effect* of the large grant, $\theta = \frac{1}{2}\beta_{LC} + \frac{1}{2}\beta_{LIK}$.

How can we perform hypothesis tests for θ ? In previous lectures, we showed that under our standard three assumptions (no perfect collinearity, IID sampling, and the orthogonality condition), OLS is asymptotically normal and the robust standard error will consistently estimate the true variance of OLS. We used this to show that $\frac{\hat{\beta} - \beta_0}{\widehat{SE}(\hat{\beta})}$ is asymptotically standard normal under the null hypothesis, which allowed us to do inference. It turns out that we can do the same thing for θ . In the multiple regressors case, not only is each coefficient normally distributed in large samples, but also the whole vector of coefficients is multivariate normal in large samples. That means that any linear combination of the coefficients will also be normally distributed in large samples, and so $\hat{\theta}$ will also be asymptotically normal. Moreover, $\frac{\hat{\theta} - \theta_0}{\widehat{SE}(\hat{\theta})}$ will be asymptotically standard normal under the null hypothesis, which will let us do inference in exactly the same way that we did for β .

Suppose that we have $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$, and we wish to test the null hypothesis $\theta = \theta_0$. To test this null hypothesis, we conduct a two-tailed test using the t-statistic, and reject if

$$\frac{|\hat{\theta} - \theta_0|}{\widehat{SE}(\hat{\theta})} > c$$

where c is selected to achieve the desired significance level. Since the t-statistic is standard normal under the null, we use the same critical values as before: for example, if we are conducting a 5% test, we use $c = 1.96$. Moreover, to construct confidence intervals, we use the formula

$$\hat{\theta} \pm c \cdot \widehat{SE}(\hat{\theta})$$

where c is the appropriate critical value (again, $c = 1.96$ for a 95% confidence interval).

Obtaining $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$

How do we obtain $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$? There are two approaches: the simple approach and the clever approach.

The Simple Approach The most straightforward way to obtain $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$ is to estimate the regression as usual, and then use the definition of θ as a linear combination of the coefficients in order to infer the

appropriate estimator and variance. Our estimate of $\hat{\theta}$ will simply be the appropriate linear combination of the $\hat{\beta}$ estimates. So, for example, if $\theta = \beta_S - \frac{1}{2}\beta_L$, then

$$\hat{\theta} = \hat{\beta}_S - \frac{1}{2}\hat{\beta}_L.$$

To get the standard error, we use the properties of the variance:

$$\begin{aligned}\widehat{SE}(\hat{\theta}) &= \sqrt{\widehat{Var}(\hat{\theta})} \\ \widehat{Var}(\hat{\theta}) &= \widehat{Var}\left(\hat{\beta}_S - \frac{1}{2}\hat{\beta}_L\right) \\ &= \widehat{Var}(\hat{\beta}_S) + \widehat{Var}\left(-\frac{1}{2}\hat{\beta}_L\right) + 2 \cdot \widehat{Cov}\left(\hat{\beta}_S, -\frac{1}{2}\hat{\beta}_L\right) \\ &= \widehat{Var}(\hat{\beta}_S) + \frac{1}{4}\widehat{Var}(\hat{\beta}_L) - \widehat{Cov}(\hat{\beta}_S, \hat{\beta}_L)\end{aligned}$$

The details will, of course, vary depending on what linear combination we are interested in, but the overall process is the same. Although we do not discuss in this class how to compute $\widehat{Cov}(\hat{\beta}_S, \hat{\beta}_L)$, Stata computes this automatically when you run a regression.

To implement this on the computer, Stata will do most of the work for you. The best way to do this is with the `lincom` command (`lincom` is short for linear combination). Suppose that you wish to run the regression

$$Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \varepsilon_i$$

and then do inference on $\theta = \beta_1 + 3 \cdot \beta_3 + 5$. In that case, you would run:

```
reg y x1 x2 x3 x4, rob
lincom x1 + 3*x3 + 5
```

which will output the estimate of $\hat{\theta}$ as well as its standard error, a 95% confidence interval, and a p-value for the null hypothesis that $\theta = 0$.⁴

The Clever Approach An alternative approach is to redefine the variables in the regression so that θ will be one of the coefficients of the transformed regression. If we do this successfully, then we can get $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$ directly from the regression. To see how we do this, let us start with the same equation as above:

$$Y_i = \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \varepsilon_i$$

First, we will manipulate this to get $(\beta_1 + 3 \cdot \beta_3)$ as one of our coefficients. The “clever” piece is figuring out what manipulations will get us our desired coefficients. If we want to get $(\beta_1 + 3 \cdot \beta_3)$ to be the new coefficient on $X_{1,i}$, we will need to add $(3 \cdot \beta_3 X_{1,i} - 3 \cdot \beta_3 X_{1,i})$ to the equation, and then distribute. We have:

$$\begin{aligned}Y_i &= \alpha + \beta_1 X_{1,i} + \beta_2 X_{2,i} + \beta_3 X_{3,i} + \beta_4 X_{4,i} + \varepsilon_i + (3 \cdot \beta_3 X_{1,i} - 3 \cdot \beta_3 X_{1,i}) \\ &= \alpha + (\beta_1 + 3 \cdot \beta_3) \cdot X_{1,i} + \beta_2 X_{2,i} + \beta_3 \cdot (X_{3,i} - 3 \cdot X_{1,i}) + \beta_4 X_{4,i} + \varepsilon_i\end{aligned}$$

Next, we can get the coefficient on $X_{1,i}$ to be $\beta_1 + 3 \cdot \beta_3 + 5 = \theta$. Adding $5 \cdot X_{1,i}$ to both sides:

$$Y_i + 5 \cdot X_{1,i} = \alpha + \underbrace{(\beta_1 + 3 \cdot \beta_3 + 5)}_{=\theta} \cdot X_{1,i} + \beta_2 X_{2,i} + \beta_3 \cdot (X_{3,i} - 3 \cdot X_{1,i}) + \beta_4 X_{4,i} + \varepsilon_i$$

So, if we run our regression with $X_{1,i}$, $X_{2,i}$, $(X_{3,i} - 3 \cdot X_{1,i})$, and $X_{4,i}$ as regressors and with $Y_i + 5 \cdot X_{1,i}$ as our new outcome, then the coefficient on $X_{1,i}$ will be θ . Note that the error term is exactly the same before and after the manipulations, and it is still uncorrelated with the transformed regressors, so we are still estimating the same underlying model. In Stata, we would implement this by running:

⁴Stata will use a t-distribution for the confidence intervals and the hypothesis test, rather than the normal distribution. This will be almost identical to the standard normal distribution, except in small samples. Of course, the standard errors and point estimate are unaffected, so `lincom` will give you everything you need to conduct the hypothesis test or construct the confidence interval yourself if you prefer to use the normal distribution.

```
gen modified_x3 = x3 - 3*x1
gen modified_y = y + 5*x1
reg modified_y x1 x2 modified_x3 x4, rob
```

which will output a regression where the coefficient on $X_{1,i}$ is θ , giving us the correct $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$.

Which approach should you use? Either method will work, and will give you the same result if done correctly. In some situations, such as when you can choose the omitted category in a regression with many dummies, it is fairly straightforward to use the “clever” approach. If you are particularly interested in a certain comparison, then defining your regressors so that the comparison of interest pops out as a coefficient can be a good idea, as long as the rest of the regression is still easy to read and interpret.

However, in general I think that simple is usually better than clever. It is easy to make a mistake in the “clever” approach, and both approaches take about the same amount of time to code.

Exercise 1. Consider the regression with cash and in-kind grants:

$$\text{Profit}_{it} = \alpha + \beta_C \text{Cash Grant}_{it} + \beta_{IK} \text{In-Kind Grant}_{it} + \gamma_t + \varepsilon_{it}$$

Define $\theta = \beta_C - \beta_{IK}$. What will $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$ be, in terms of the estimated coefficients and their variance and covariances? How would you construct a test for the null hypothesis $\theta = 0$? How would you construct a confidence interval?

Exercise 2. Continuing with the cash vs. in-kind regression, how would things change if you instead defined $\theta = 2 \cdot \beta_C - 2 \cdot \beta_{IK}$? How would $\hat{\theta}$ and $\widehat{SE}(\hat{\theta})$ change? Would the t-statistic change?

Exercise 3. Continuing with the cash vs. in-kind regression, show that if we replace $\text{In-Kind Grant}_{it}$ with $(\text{Cash Grant}_{it} + \text{In-Kind Grant}_{it})$, then we will obtain the regression:

$$\text{Profit}_{it} = \alpha + \theta \cdot \text{Cash Grant}_{it} + \beta_{IK} (\text{Cash Grant}_{it} + \text{In-Kind Grant}_{it}) + \gamma_t + \varepsilon_{it}$$

How should we interpret the variable $(\text{Cash Grant}_{it} + \text{In-Kind Grant}_{it})$? Why might it make sense to just call it Grant_{it} ?

Exercise 4. Continuing with the cash vs. in-kind regression, show that if we also replace Cash Grant_{it} with $\left(\frac{1}{2}\text{Cash Grant}_{it} - \frac{1}{2}\text{In-Kind Grant}_{it}\right)$, then we will obtain the regression:

$$\text{Profit}_{it} = \alpha + \theta \cdot \left(\frac{1}{2}\text{Cash Grant}_{it} - \frac{1}{2}\text{In-Kind Grant}_{it}\right) + \delta \cdot (\text{Cash Grant}_{it} + \text{In-Kind Grant}_{it}) + \gamma_t + \varepsilon_{it}$$

where $\delta = \frac{1}{2}(\beta_C + \beta_{IK})$ is the “average effect” of the two types of grants. *Hint: Rather than doing more manipulations, you may find it easier to think about the expected value of Profit_{it} under the control, cash grant, and in-kind grant treatments. It is sufficient here to show that the above regression will give exactly the same predictions as the original regression.*

Exercise 5. Consider the regression with small and large grants:

$$\text{Profit}_{it} = \alpha + \beta_L \text{Large Grant}_{it} + \beta_S \text{Small Grant}_{it} + \gamma_t + \varepsilon_{it}$$

Define $\theta = \beta_S - \frac{1}{2}\beta_L$. Show that if we replace Large Grant_{it} with $\left(\text{Large Grant}_{it} + \frac{1}{2} \cdot \text{Small Grant}_{it}\right)$, then we will obtain the regression:

$$\text{Profit}_{it} = \alpha + \beta_L \left(\text{Large Grant}_{it} + \frac{1}{2} \cdot \text{Small Grant}_{it}\right) + \theta \cdot \text{Small Grant}_{it} + \gamma_t + \varepsilon_{it}$$